These Outflows are (Not) the Quenching Mechanisms You're Looking For

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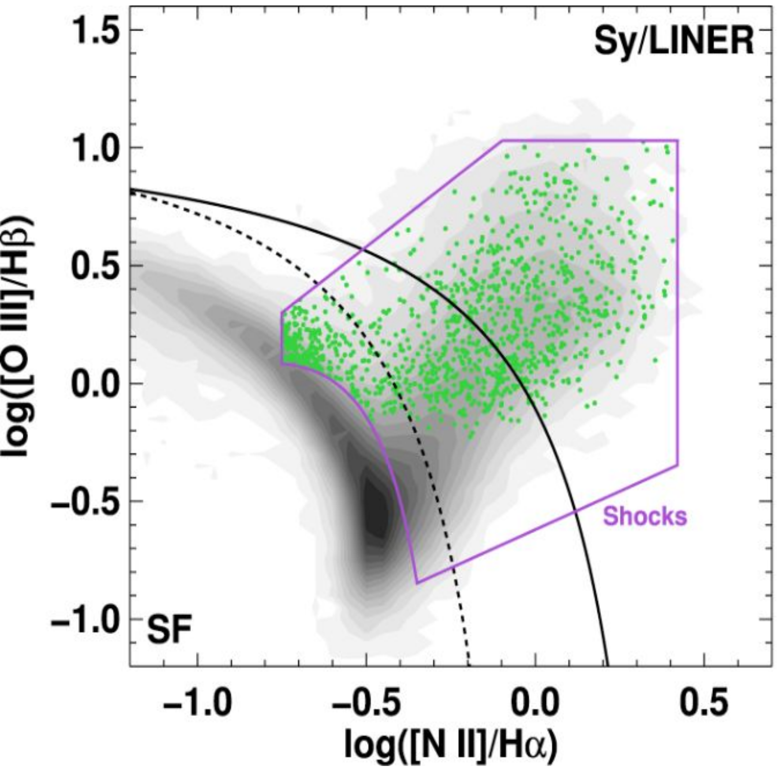
Taylor Tomko

In collaboration with the
post-starburst group at UIUC
and Space Telescope

The Shocked POst-starburst Galaxy Survey



Selection criteria:

1. Galaxies within shock boundary
2. EW Hδ > 5Å

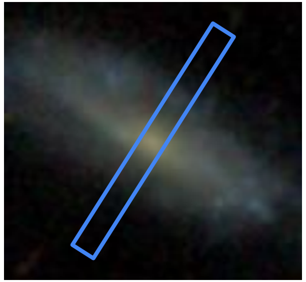
1067 total SPOGs with nuclear fiber data

Alatalo+16

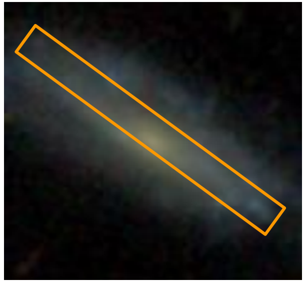
SDSS DR7

Post-starbursts (might) rapidly quench, but:

★ Does ionized gas in SPOGs get removed by outflows?



★ How does the ionized gas in the ISM look?



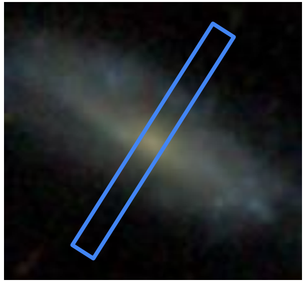
SPOGs Observations with the LDT

SPOGs Target Selection:

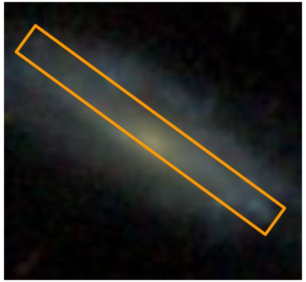
Observed a sample of 111

1. Edge-on
2. Low redshift (< 0.1)
3. Brighter in sample (g-magnitude 15-17)

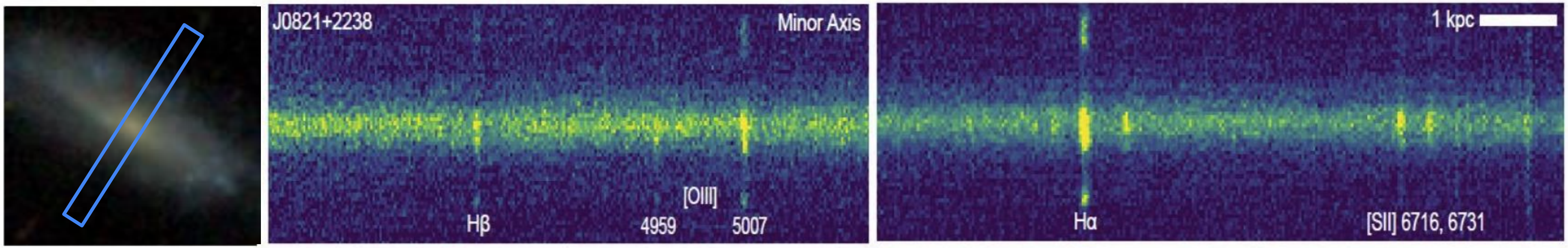
Minor Axis



Major Axis



SDSS DR7



Project 1: SPOGs Minor Axis Spectroscopy

arXiv > astro-ph > arXiv:2412.06621

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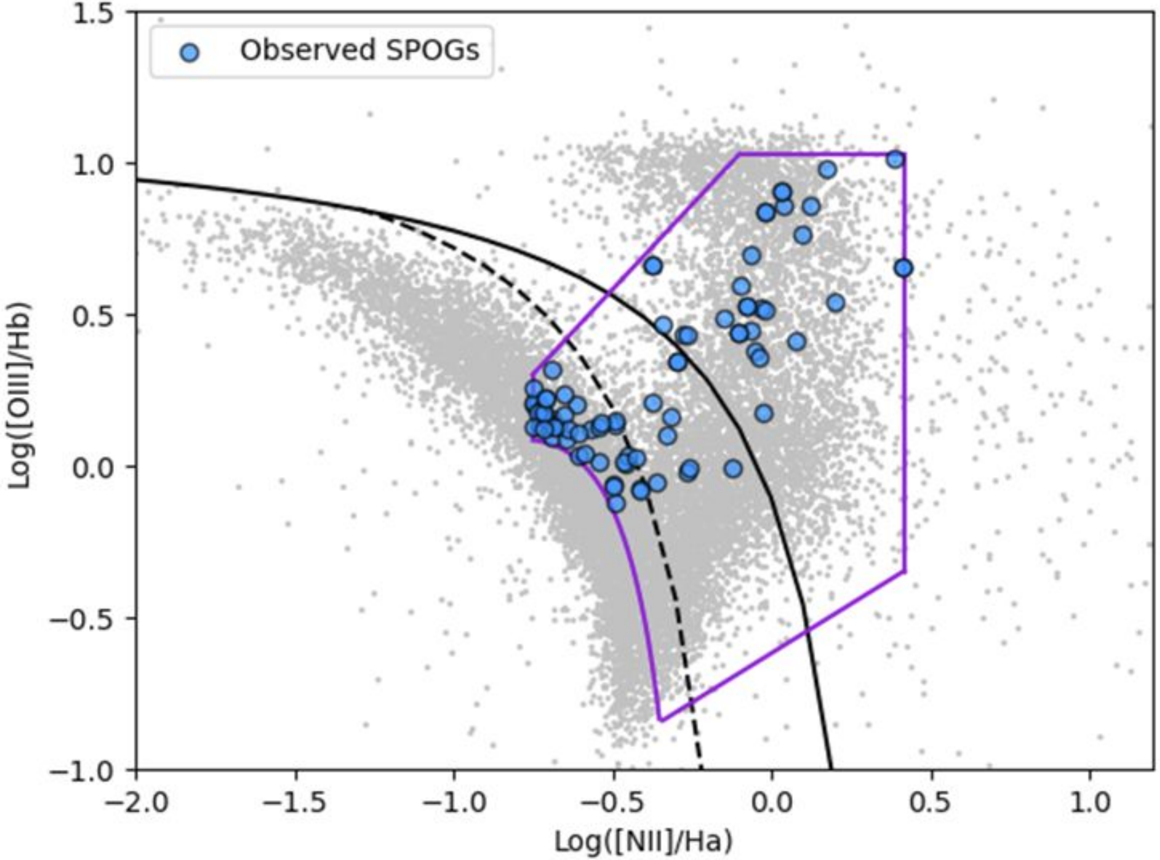
Astrophysics > Astrophysics of Galaxies

[Submitted on 9 Dec 2024]

Shocked POststarburst Galaxy Survey. IV. Outflows in Shocked Post-Starburst Galaxies Are Not Responsible For Quenching

Antoniu Fodor, Taylor Tomko, Mary Braun, Anne M. Medling, Thomas M. Johnson, Alexander Thompson, Victor D. Johnston, Matthew Newhouse, Yuanze Luo, K. Decker French, Justin A. Otter, Akshat Tripathi, Margaret E. Verrico, Katherine Alatalo, Kate Rowlands, Timothy Heckman

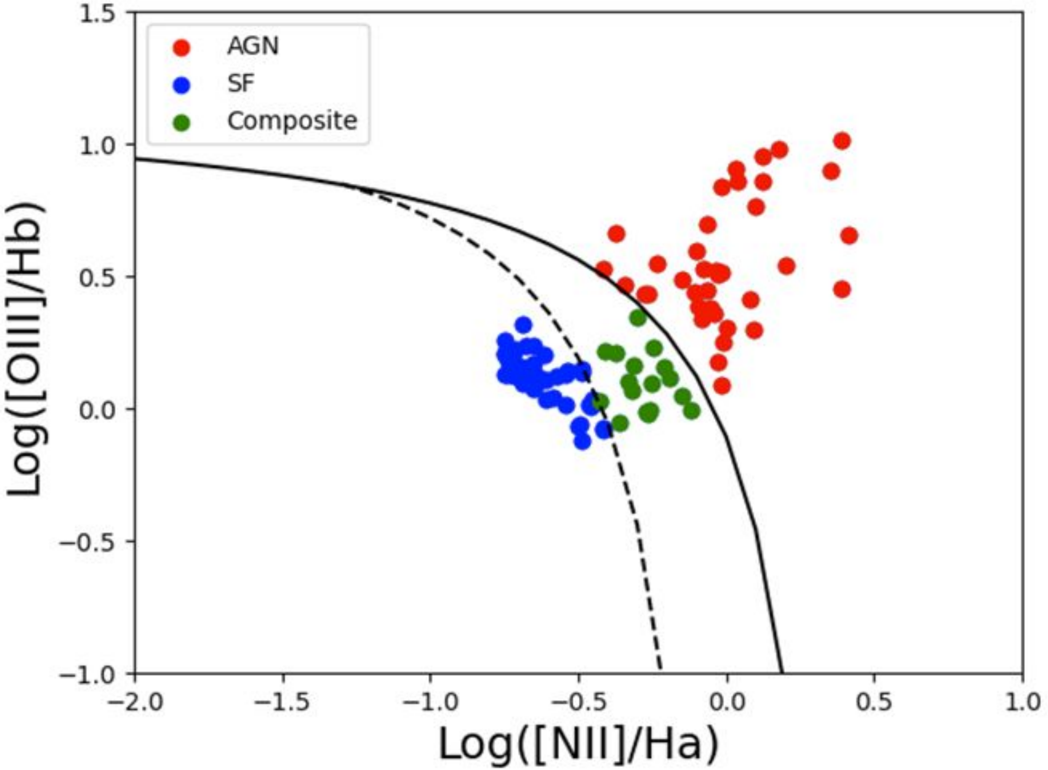


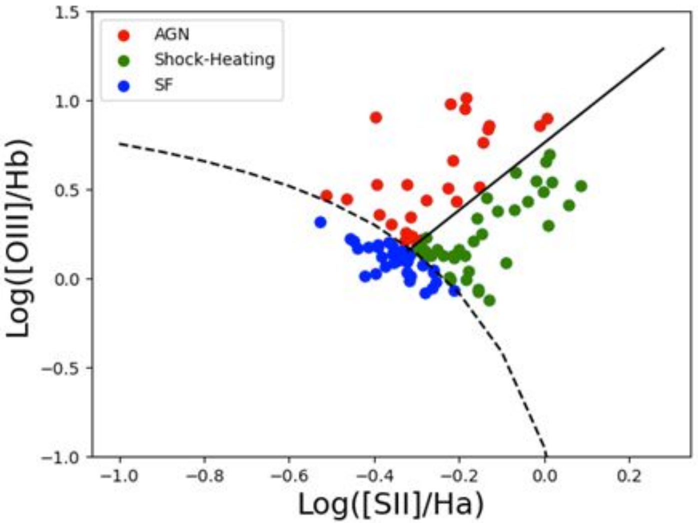
★ 77 SPOGs have b/a < 0.7

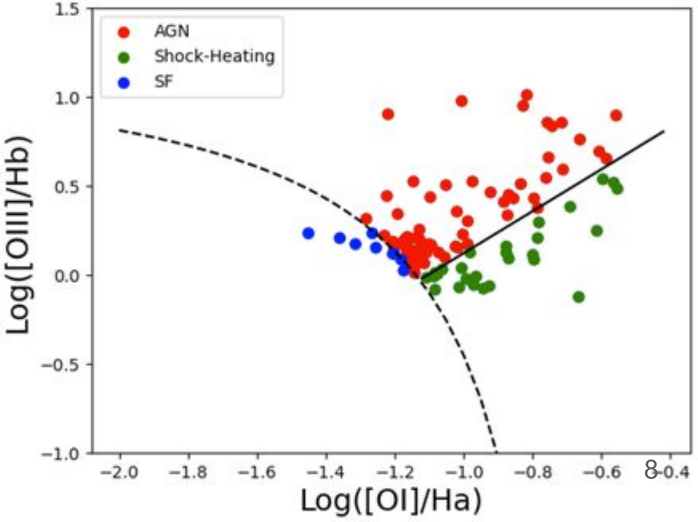
threshold set for minor axis gas detections

SPOGs split into subsamples based on nuclear line ratio classification:

-AGN SPOGs
-SF SPOGs
-LINER SPOGs

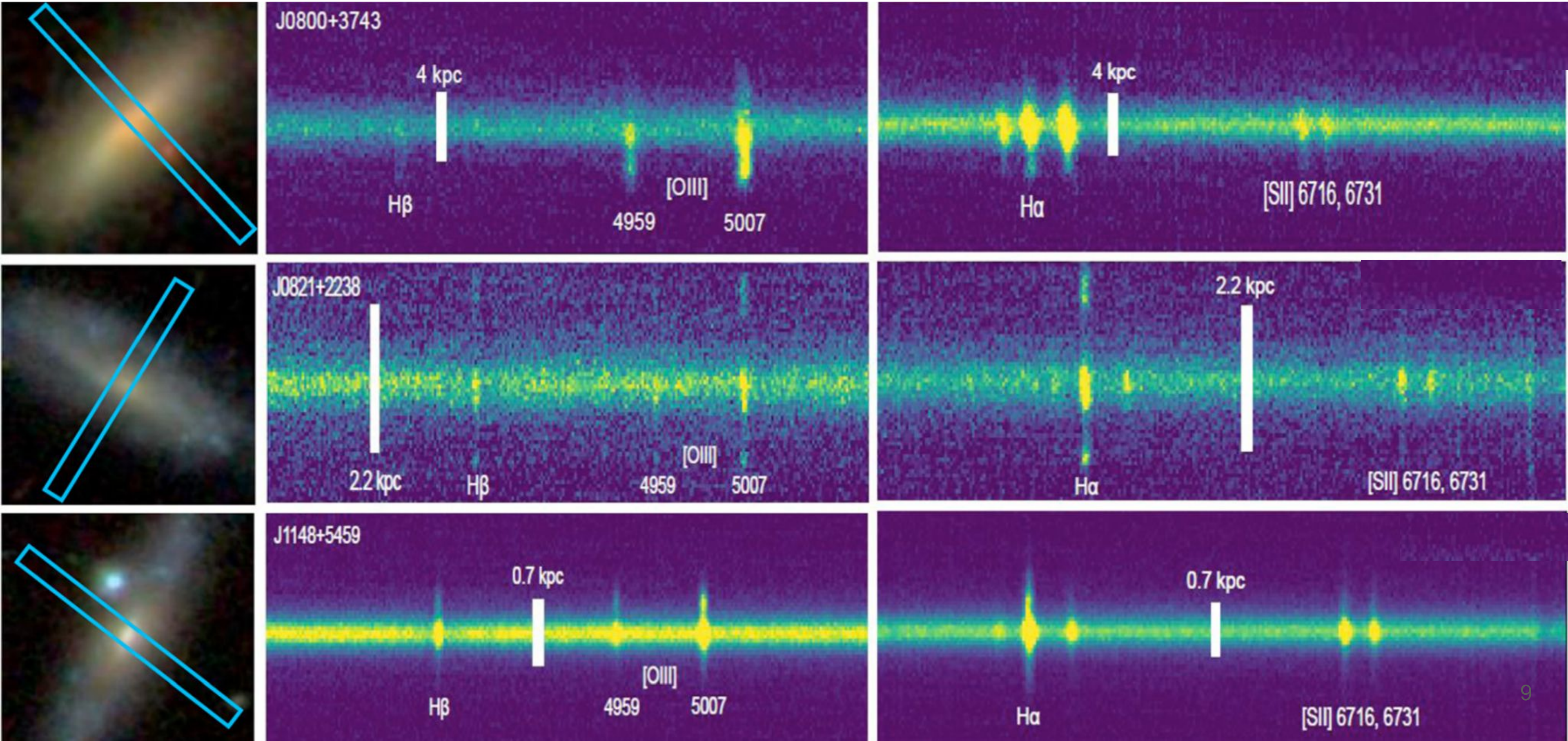






Do we see extraplanar gas? Yes!

Fodor+25



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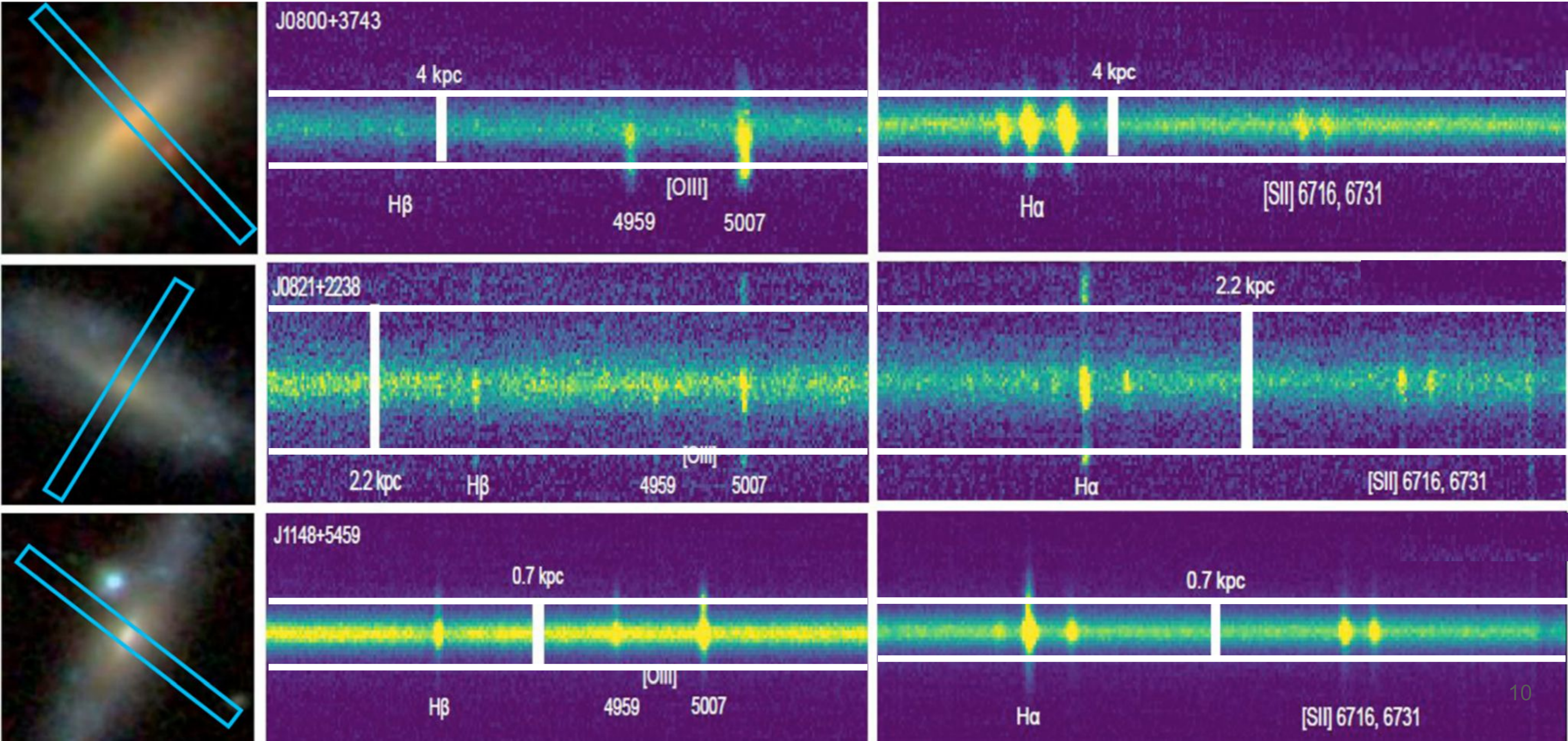

40% of SPOGs observed contain detected extraplanar gas

Galaxy Classifications	Count	Tot. Extraplanar Gas Detections	Tot. Extraplanar Fraction	Hα Count	Hα Fraction	[O III] Count	[O III] Fraction
(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Total	77	31	40.3%	29	37.7%	15	19.5%
AGN	23	12	52.2%	10	43.5%	10	43.5%
LINER	20	4	20%	4	20%	1	5%
SF	34	15	44.1%	15	44.1%	4	11.8%

Majority of [OIII] detection in AGN SPOGs

Galaxy Classifications (1)	Count (2)	Tot. Extraplanar Gas Detections (3)	Tot. Extraplanar Fraction (4)	Hα Count (5)	Hα Fraction (6)	[O III] Count (7)	[O III] Fraction (8)
Total	77	31	40.3%	29	37.7%	15	19.5%
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Is all extraplanar gas outflowing?

Image credit: *J. Tumlinson, M.S. Peeples and J.K. Werk/*Annual Review of Astronomy and Astrophysics *2017; M.S. Peeples/*Nature *2015*

Is all extraplanar gas outflowing? No!

★ eDIG - diffuse ionized gas around a galaxy
  - $\sigma_{gas} < 60$ km/s

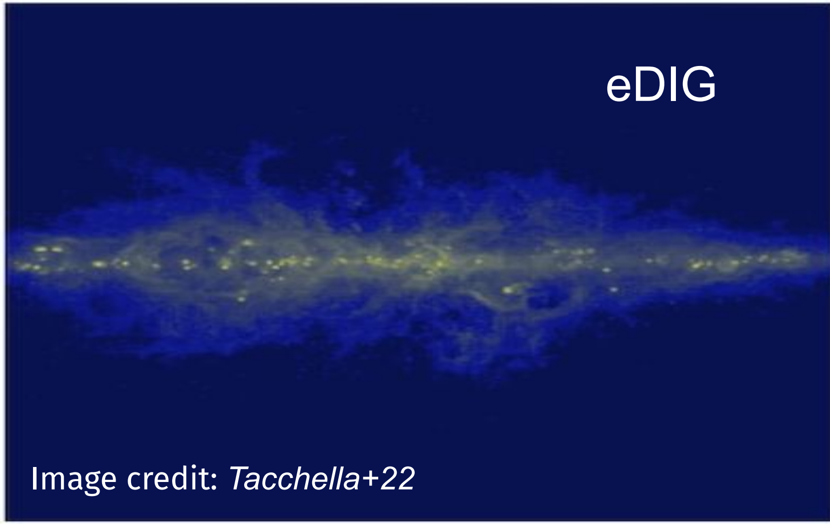
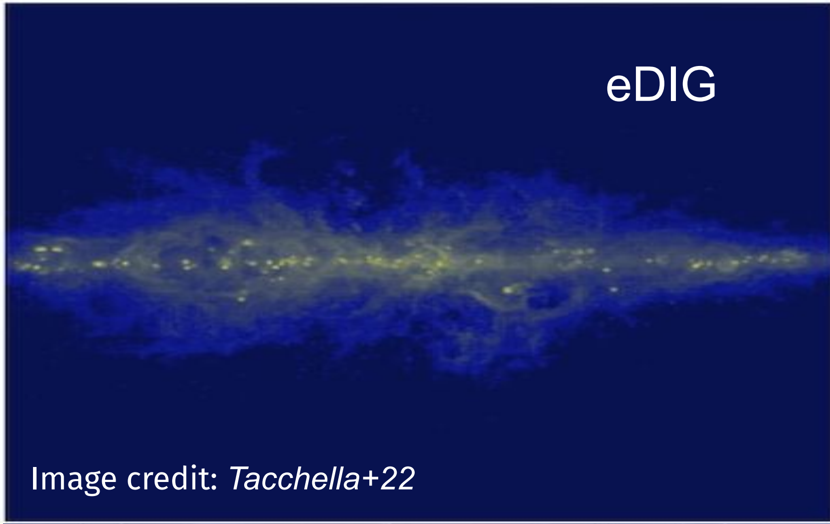
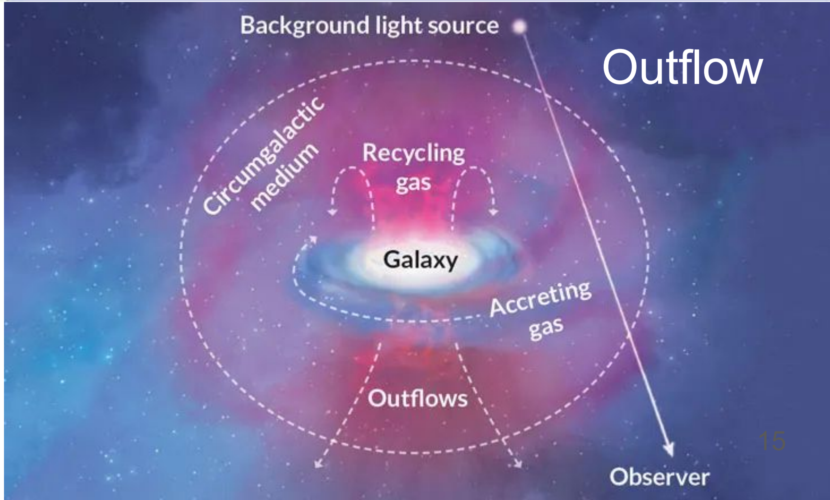


Image credit: *J. Tumlinson, M.S. Peeples and J.K. Werk/*Annual Review of Astronomy and Astrophysics *2017; M.S. Peeples/*Nature *2015*

Is all extraplanar gas outflowing?
No!

- ★ eDIG - diffuse ionized gas around a galaxy
  - ○ $\sigma_{gas} < 60$ km/s
- ★ Weak outflow - potentially has gas being removed, but errors make it difficult to tell
  - ○ $\sigma_{external} > 60$ km/s but less than $\sigma_{internal}$ + error
- ★ Outflowing gas - gas is being removed from a galaxy
  - ○ $\sigma_{external} > \sigma_{internal}$ + error





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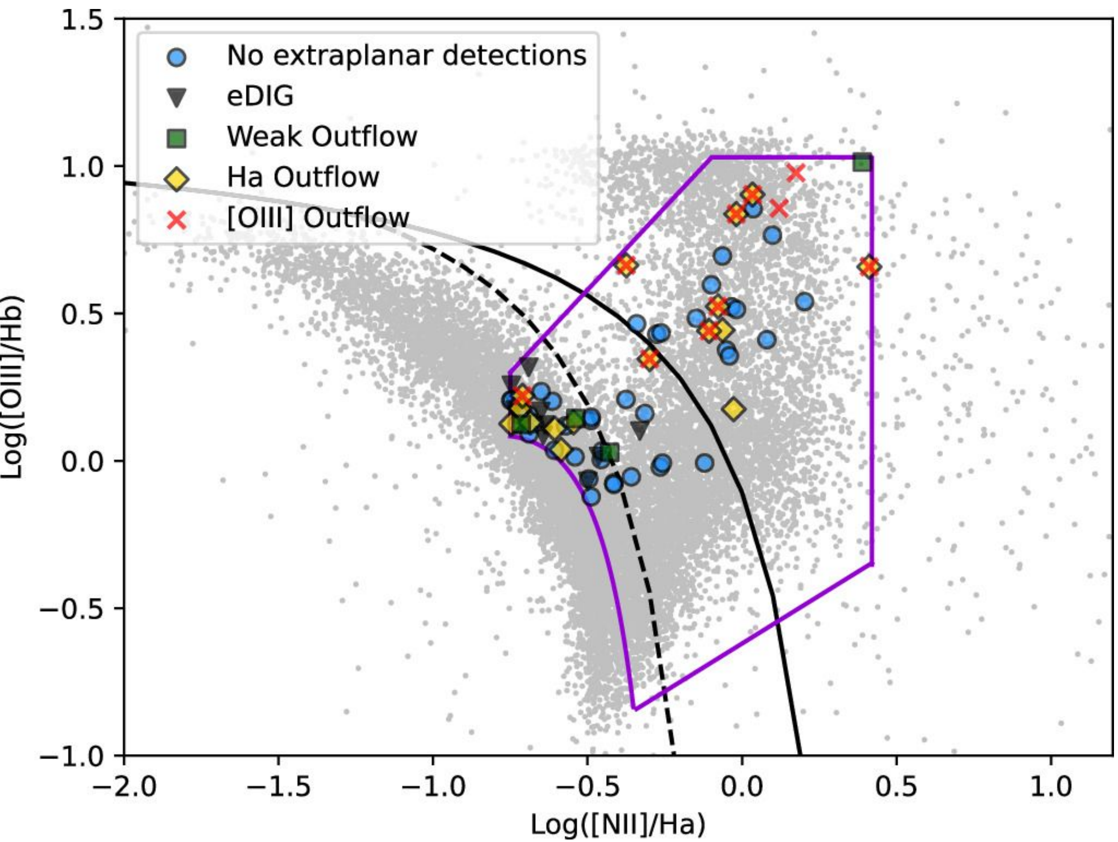

- ★ Outflowing extraplanar gas
  - 18 SPOGs
- ★ Weakly outflowing extraplanar gas
  - 4 SPOGs
- ★ eDIG
  - 9 SPOGs (LINER+SF SPOGs)

Fodor+25

SPOGs do not have significant masses entrained in outflows

Galaxy name (J2000) (1)	Hα extent (kpc) (2)	[O III] extent (kpc) (3)	Internal σ (km s^{-1}) (4)	External σ (km s^{-1}) (5)	Extraplanar Mass ($M_\odot$) (6)	Mass Loading Factor (7)
AGN						
J0004-0114	6.2	4.5	153	169	6.3×10^5	3×10^{-3}
J0102-0052	2.7	2.7	132	136	2×10^4	4×10^{-3}
J0755+3929	2.9	2.9	225	244	3.7×10^5	2×10^{-3}
SF						
J0141-0015 †	2.2	—	40	26	†	†
J0204+0051 ^{W}	5.9	—	64	78	1×10^5	1.2×10^{-5}
J0728+3654 †	2.4	—	93	67	†	†
J0813+2434 ^{W}	1.1	—	92	103	2×10^3	2.4×10^{-6}

Adapted from Fodor+25

SPOGs do not have efficient outflows

Galaxy name (J2000) (1)	Hα extent (kpc) (2)	[O III] extent (kpc) (3)	Internal σ (km s^{-1}) (4)	External σ (km s^{-1}) (5)	Extraplanar Mass ($M_\odot$) (6)	Mass Loading Factor (7)
AGN						
J0004-0114	6.2	4.5	153	169	6.3×10^5	3×10^{-3}
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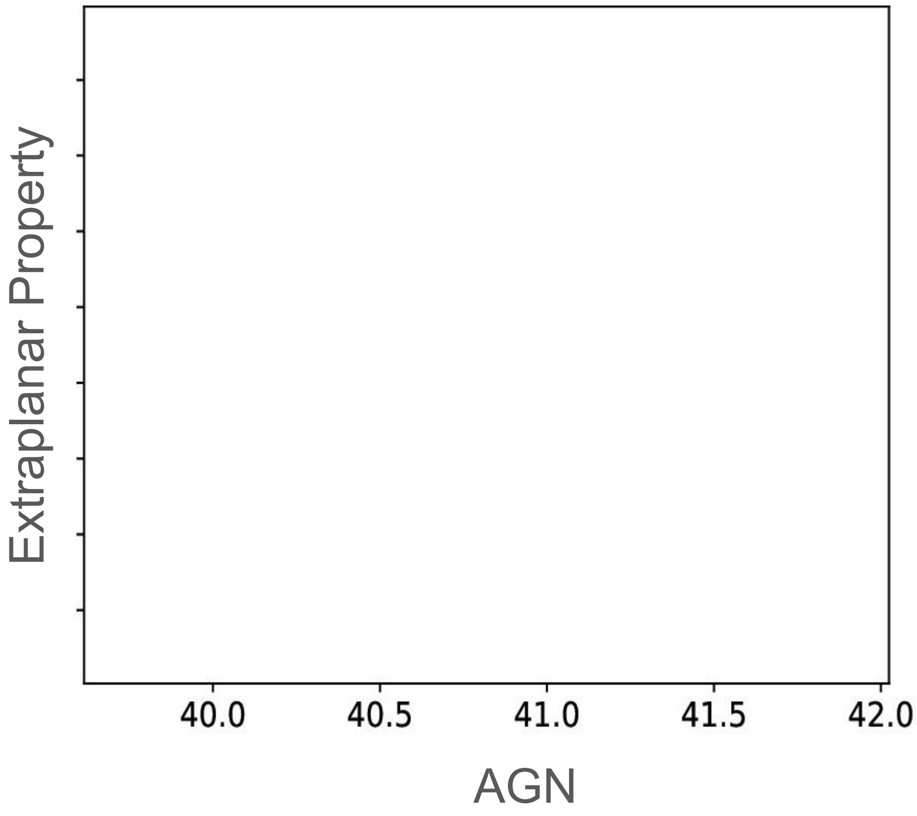
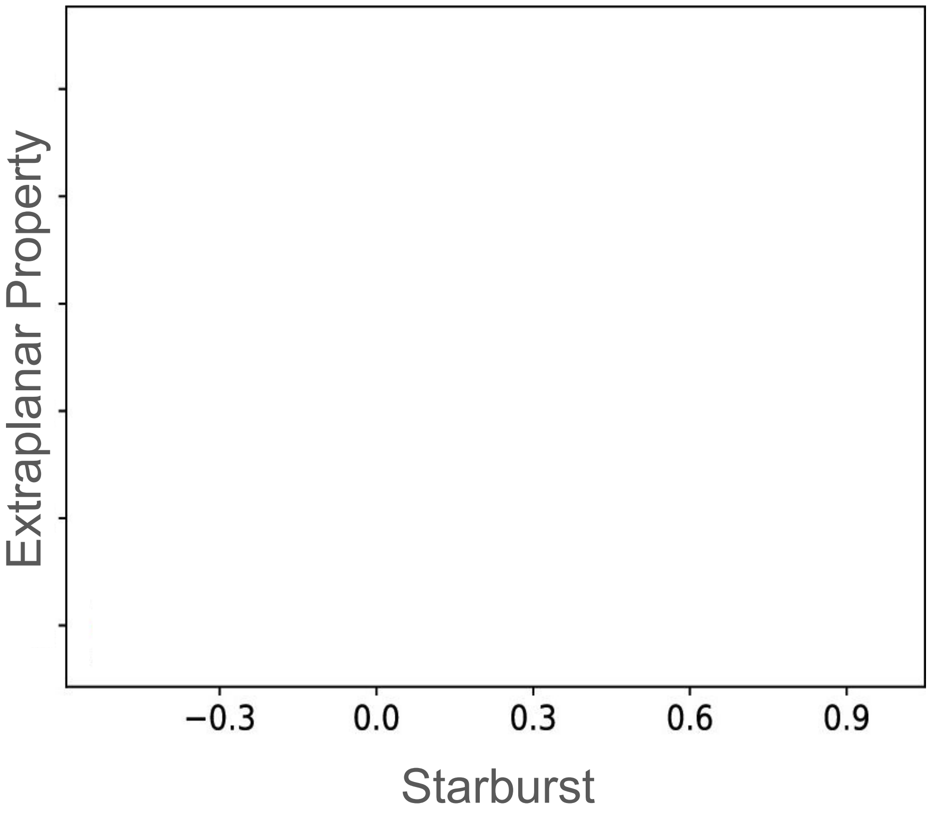
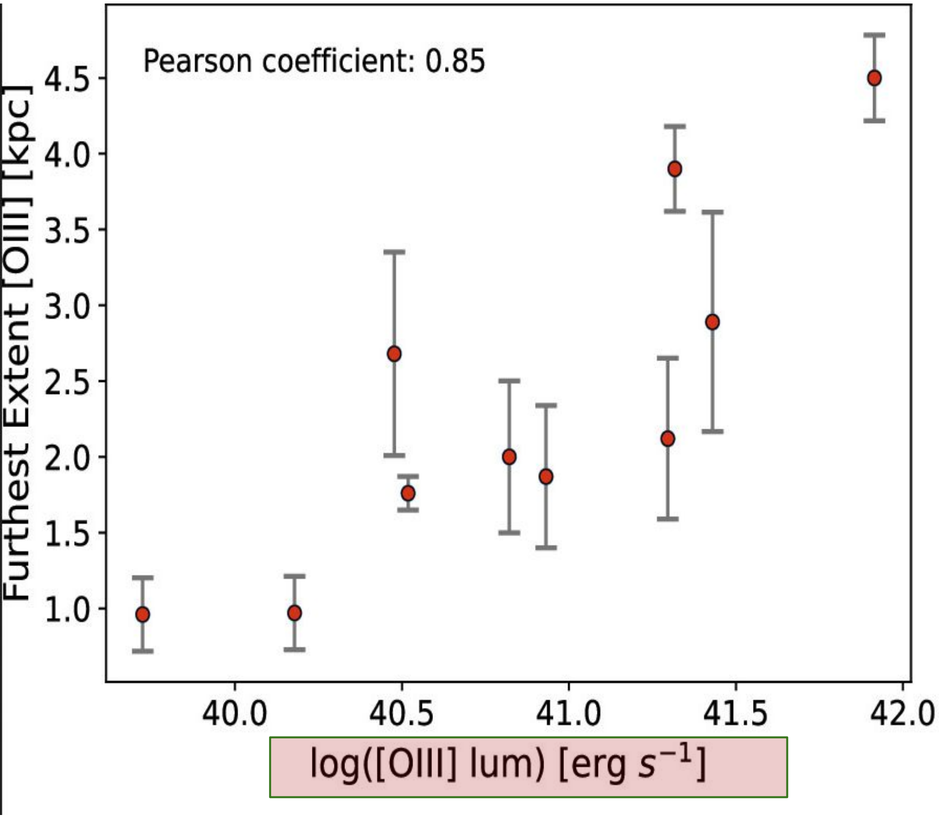
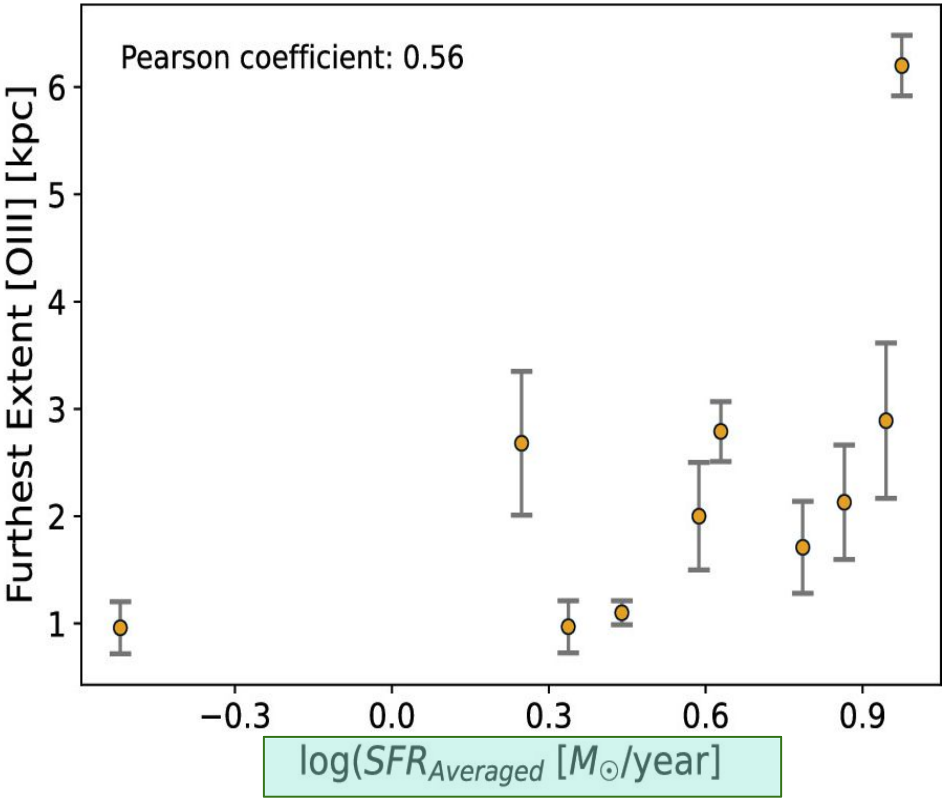
AGN SPOGs- What physical mechanisms drive the outflows?





AGN SPOGs- What physical mechanisms drive the outflows?





Fodor+25

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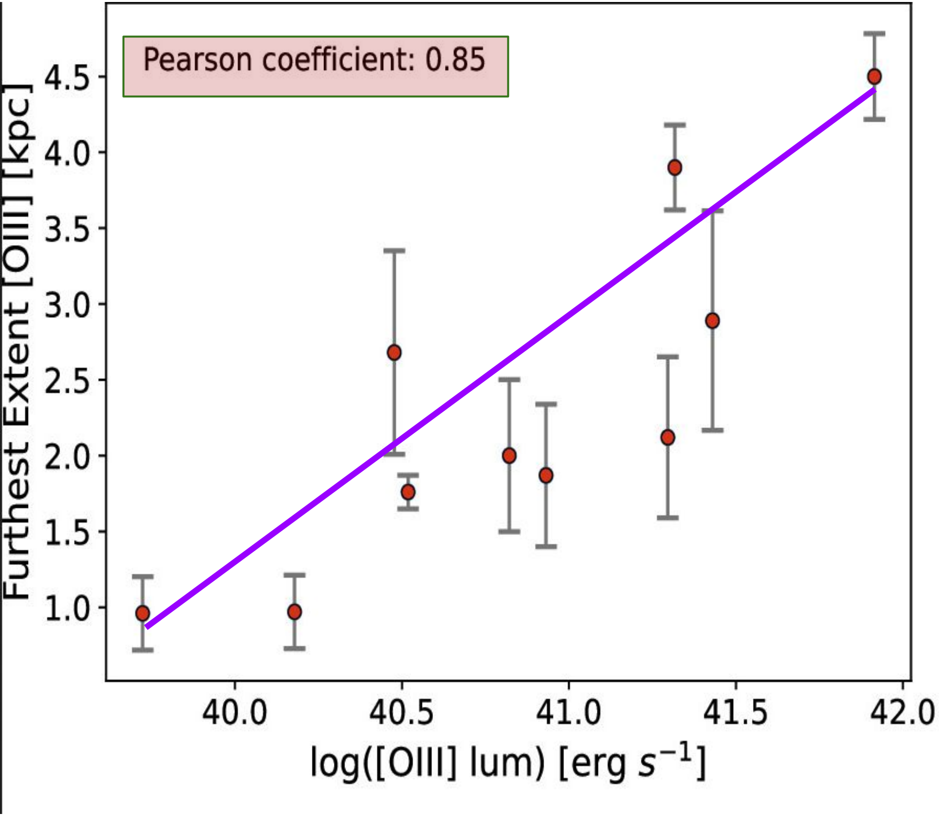

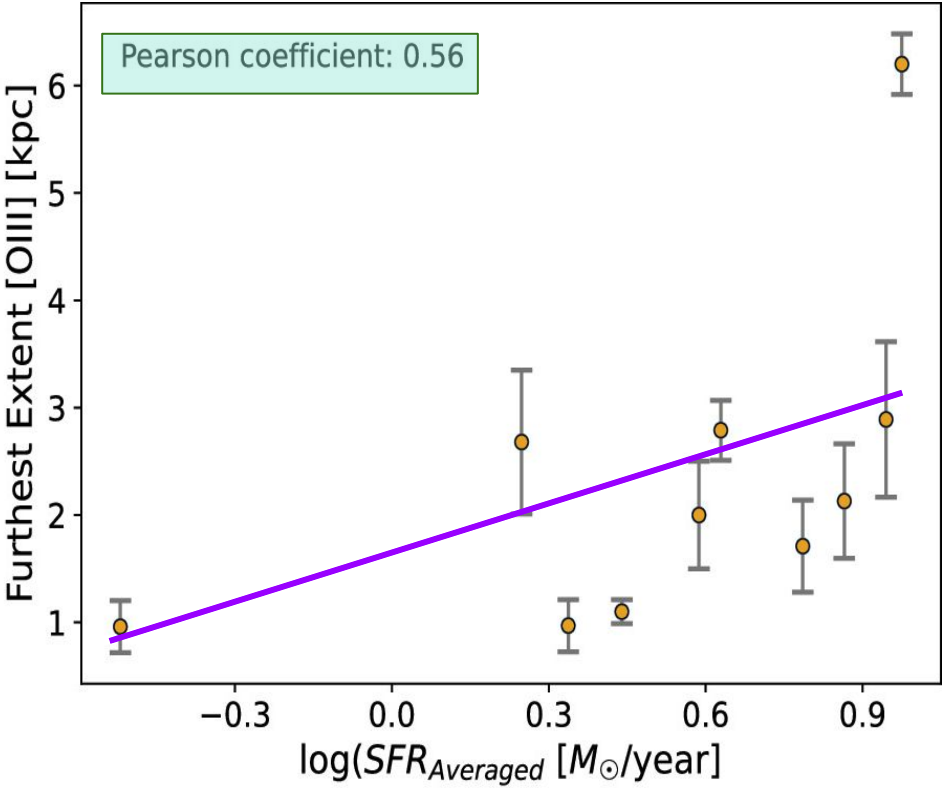

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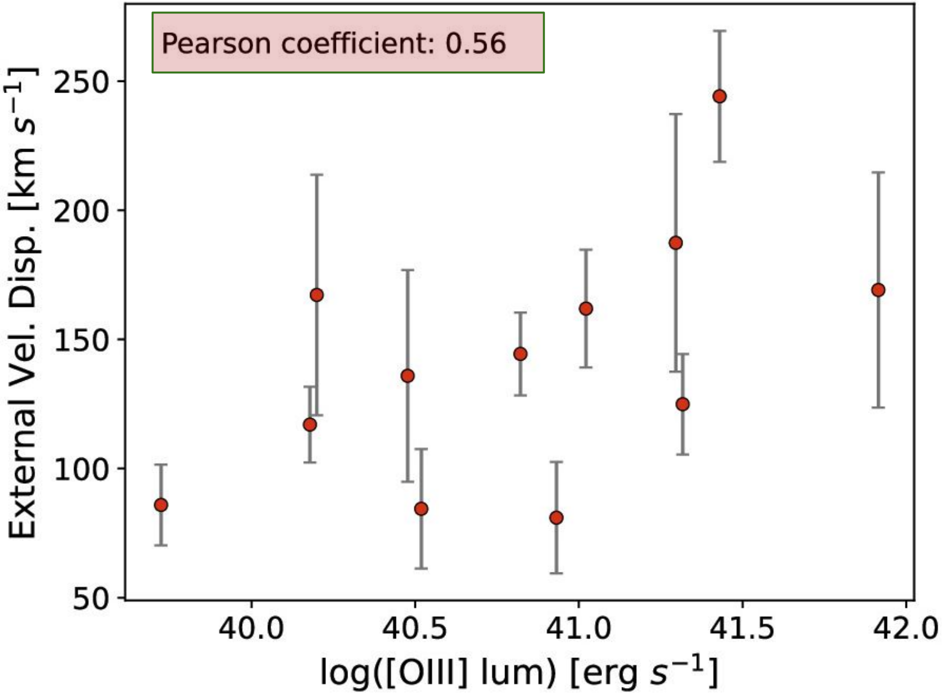

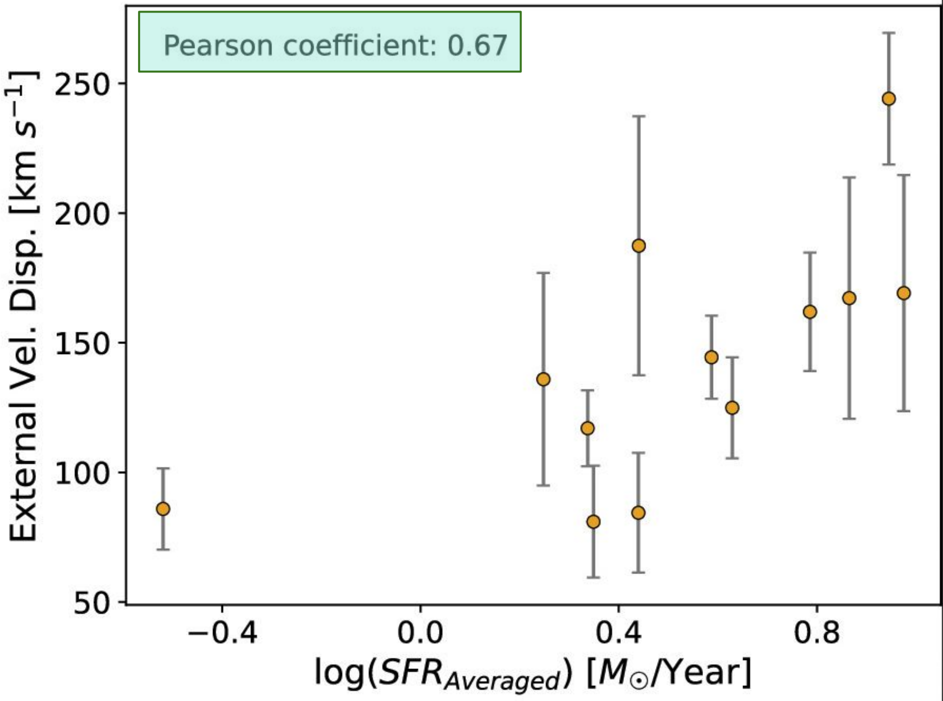

Fodor+25

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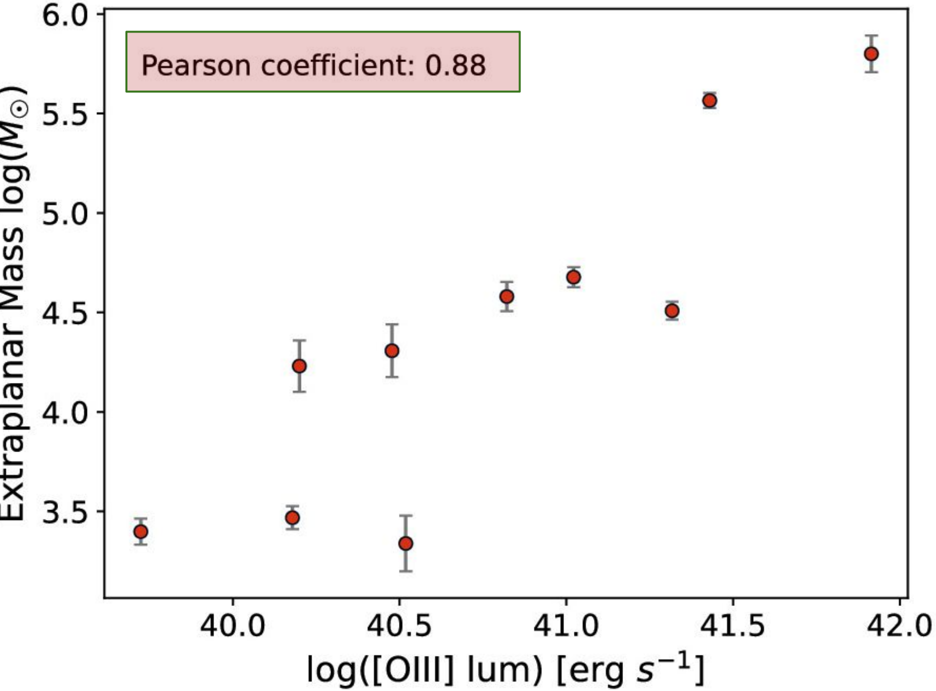

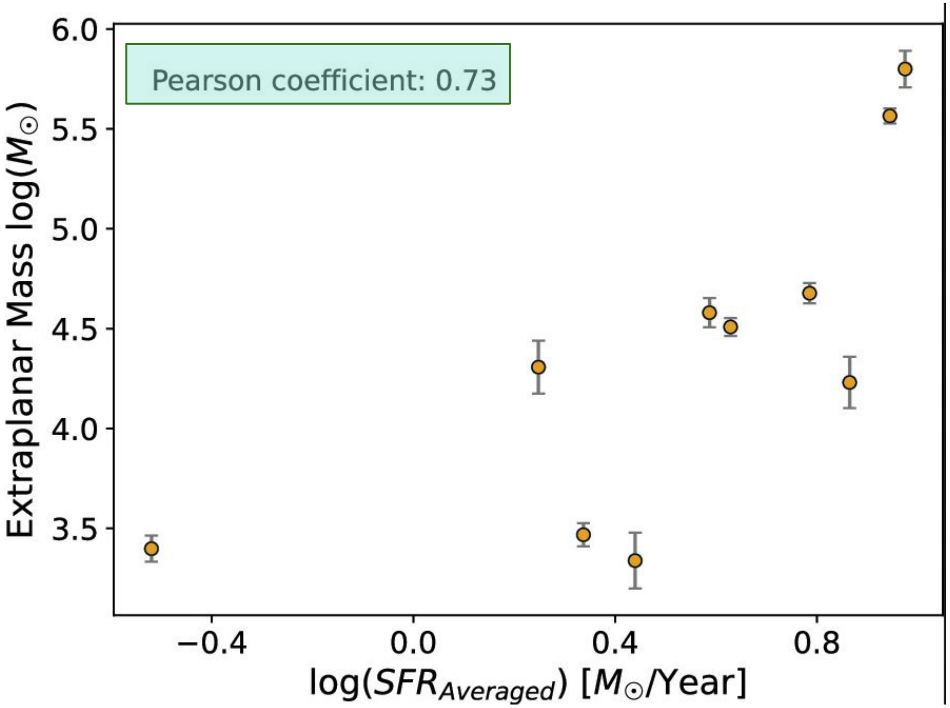

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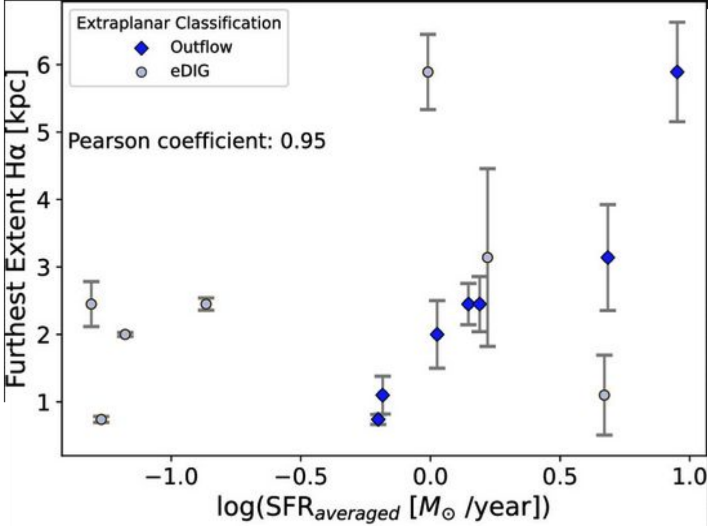
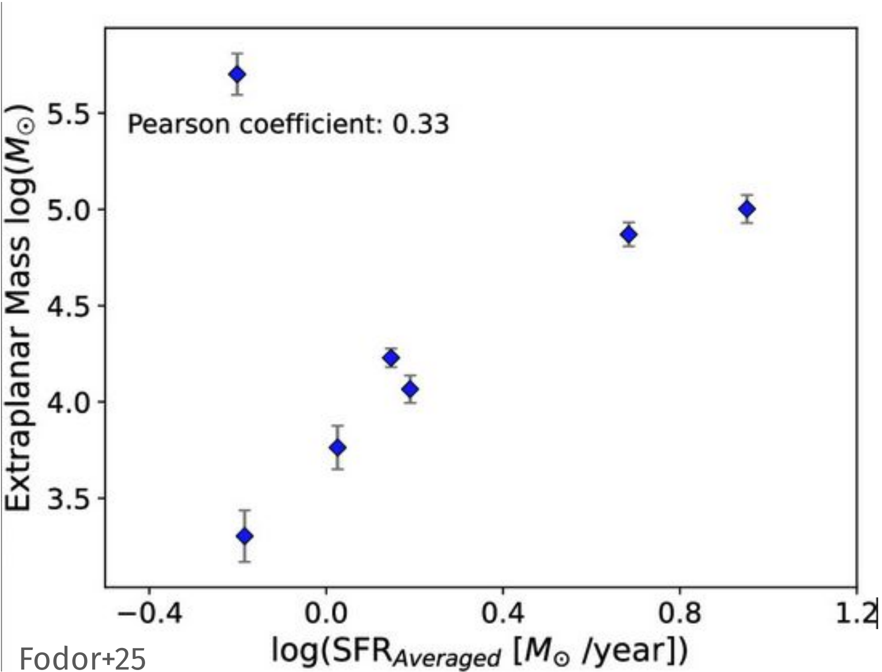
AGN SPOGs Outflow Driving Mechanism?

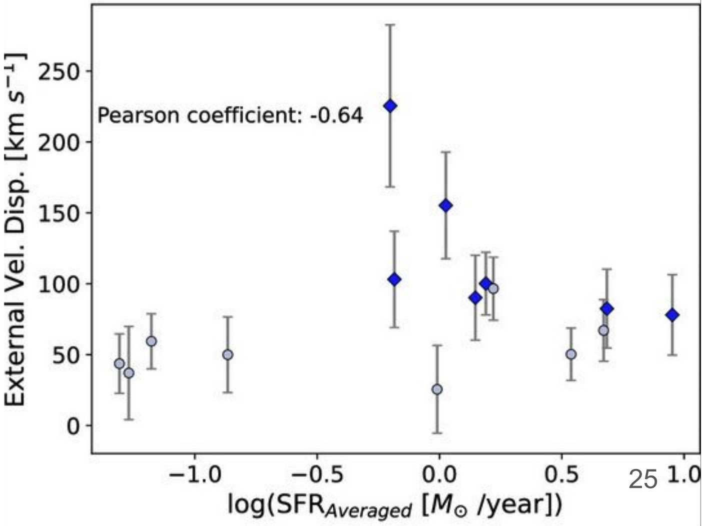
- ★ Furthest extent: likely AGN
- ★ Extraplanar Velocity Dispersion: either AGN or stellar feedback
- ★ Extraplanar gas mass: likely AGN

$\boxed{\text{Thus, AGN SPOGs have outflows driven by the central AGN}}$

For SF SPOGs, strong trend with $SFR_{Averaged}$ for 2 extraplanar properties

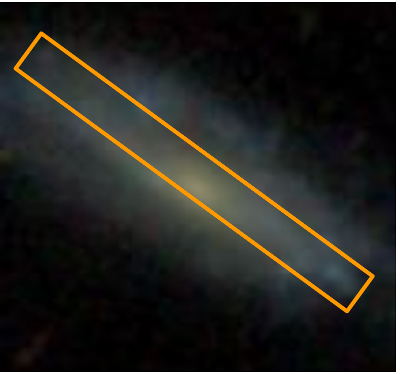


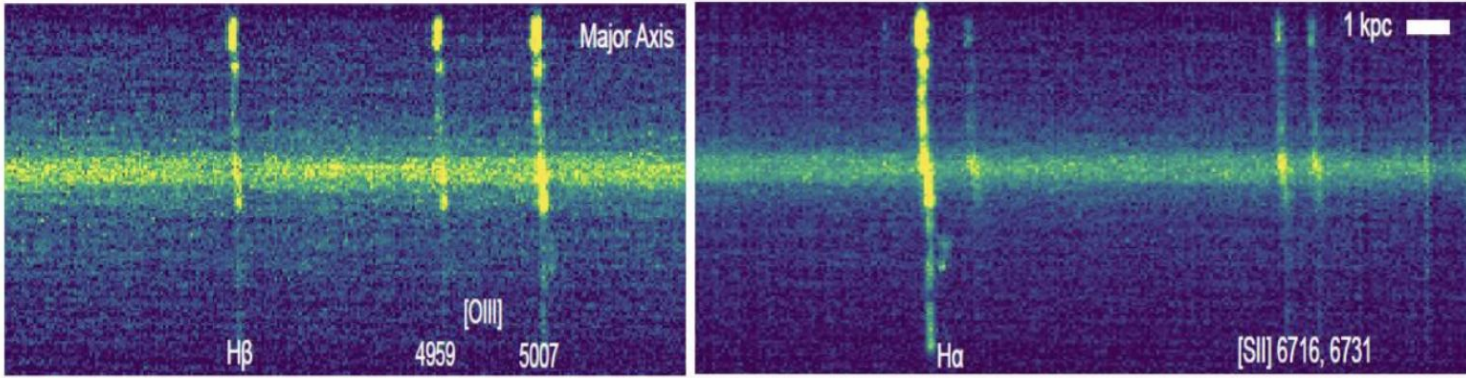




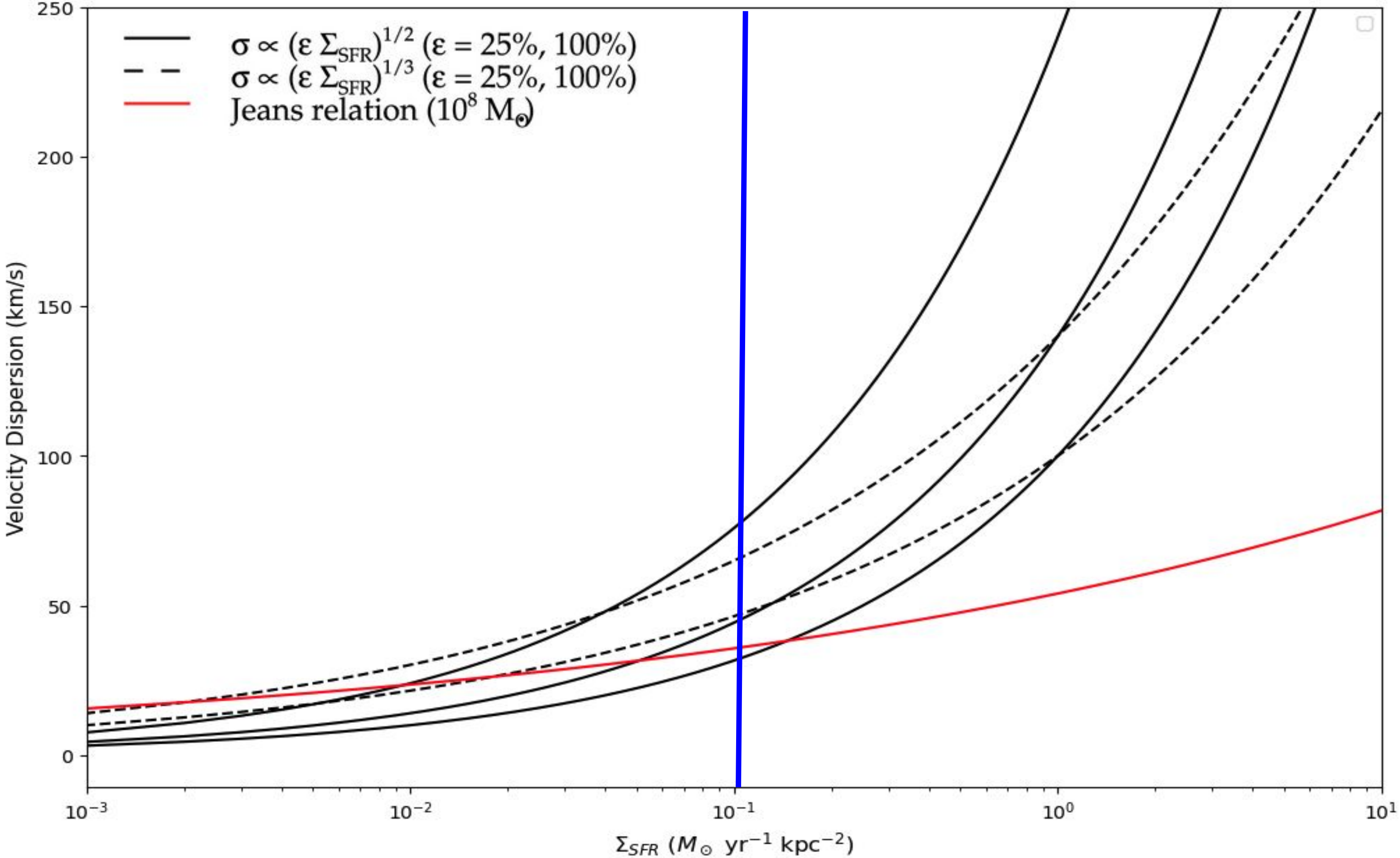
Fodor+25

Project 2: SPOGs Major Axis Spectroscopy



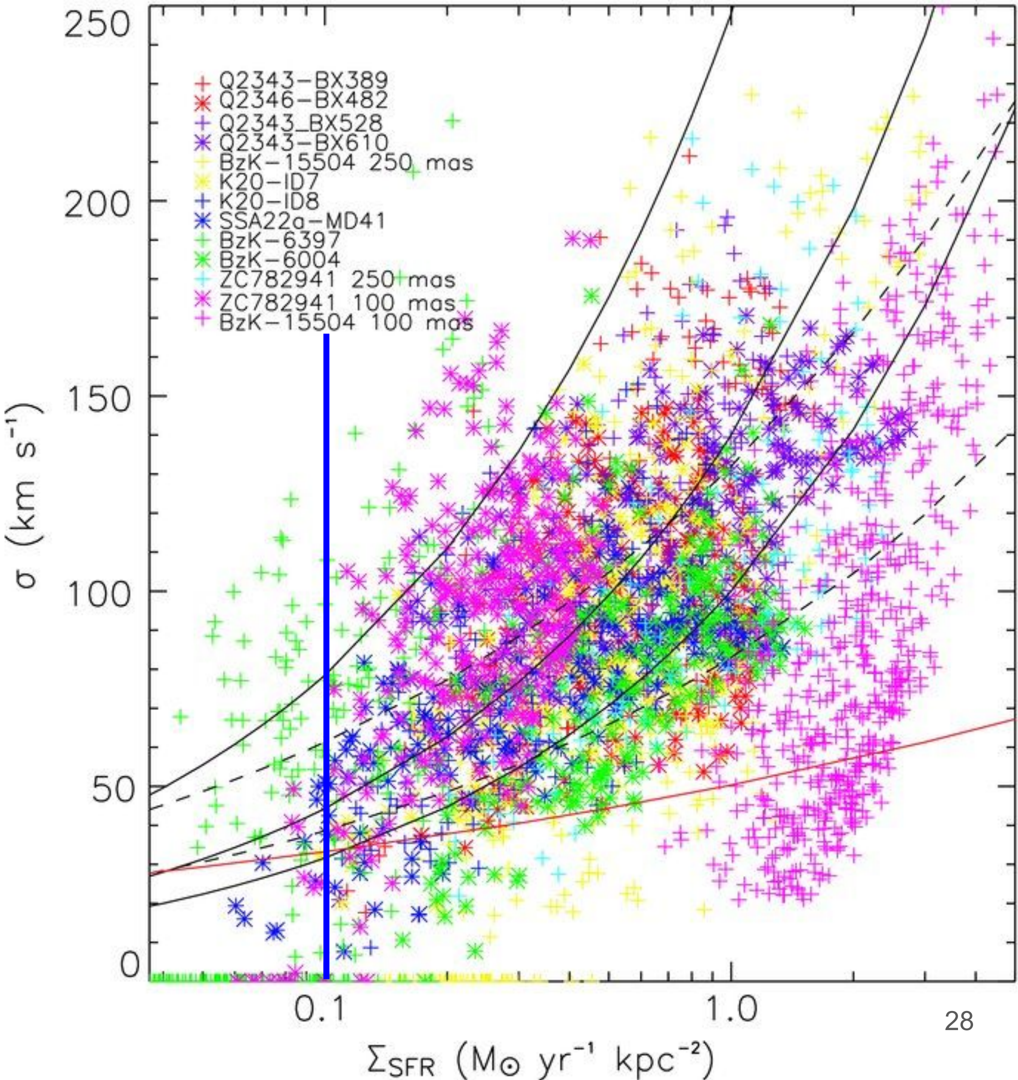


Predictions of turbulence based only on stellar feedback

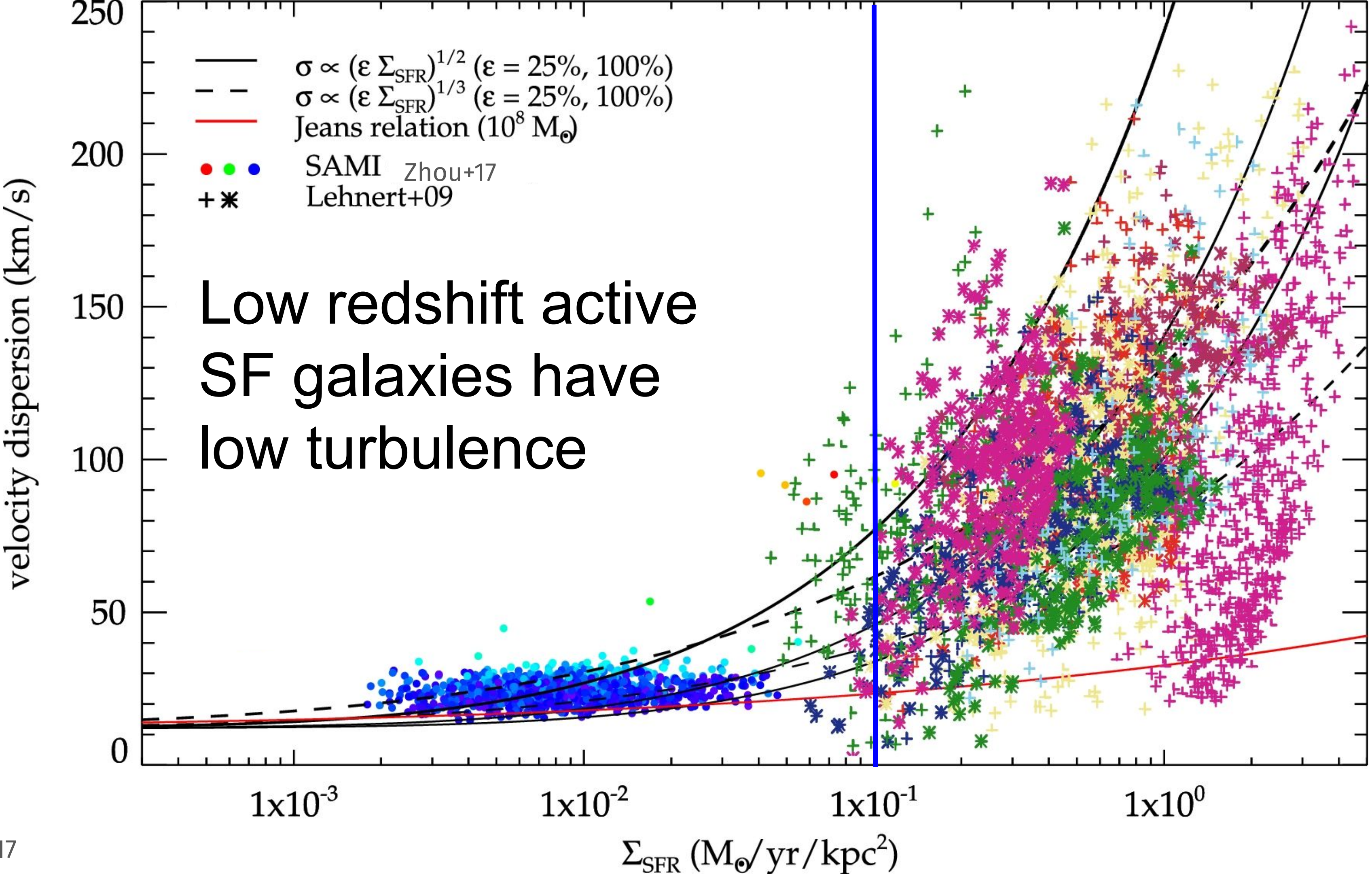


Lehnert+09

Active SF galaxies (~z=2) have a range of velocity dispersion (turbulence)



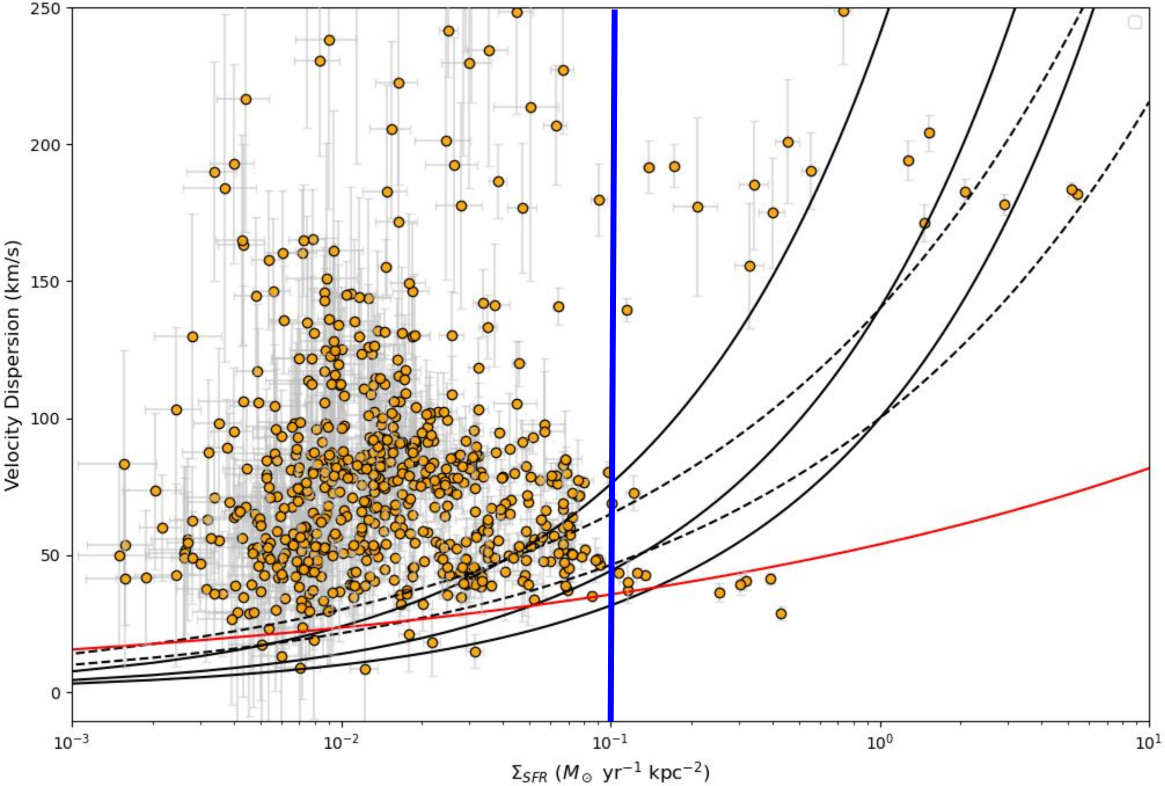
Lehnert+09



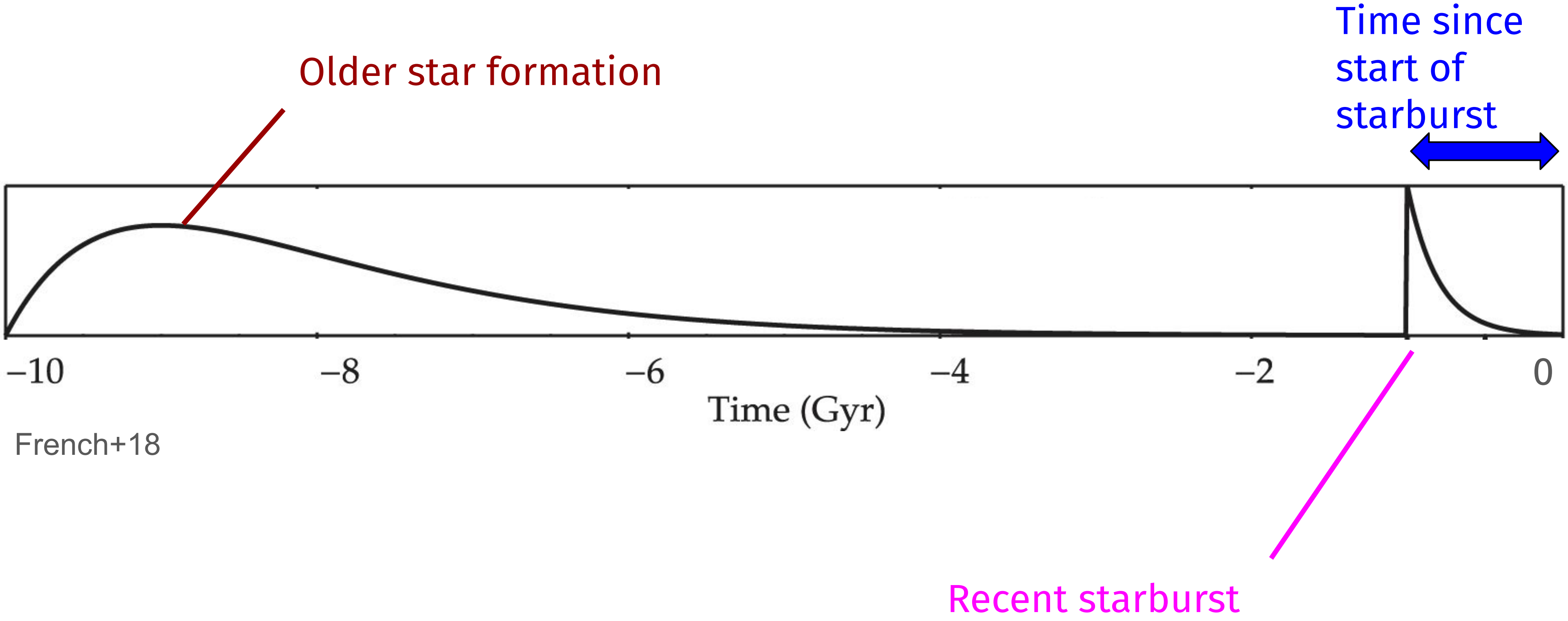
Low redshift active SF galaxies have low turbulence

Zhou+17

SF spaxels in SPOGs have higher velocity dispersion than expected from stellar feedback models and normal galaxies

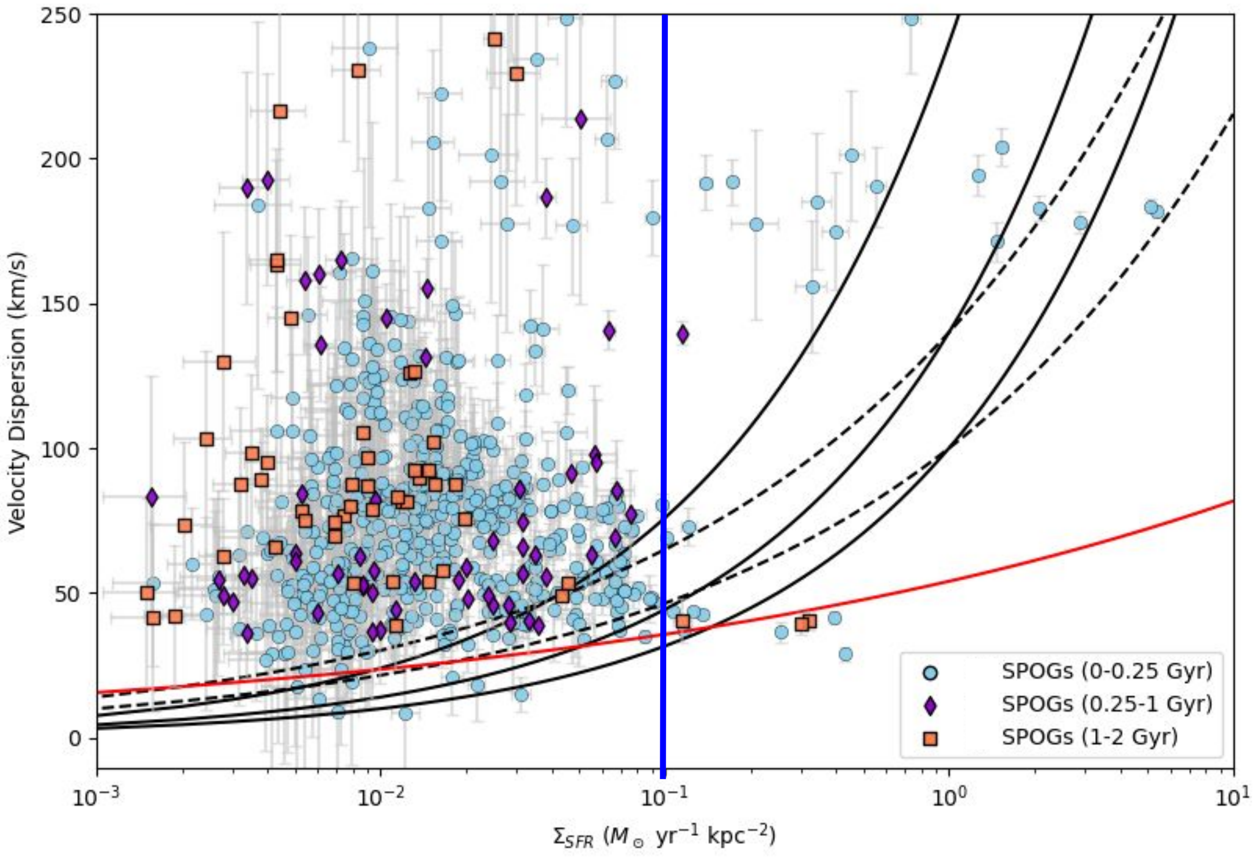


Modeling recent star formation histories for post-starbursts and SPOGs



French+18

Gas stays turbulent for at least 1-2 billion years after the burst starts



Fodor et al. in prep

Summary

- Do we see extraplanar gas? Yes! In 40% of SPOGs
- Is the gas outflowing? Yes! In 29% of SPOGs, mostly AGN SPOGs
- Is there a substantial amount of mass entrained in the outflow? No, not driving much gas out. Outflows are not quenching SF

- How do SPOGs compare to normal SF galaxies? SPOGs contain much more turbulent gas at similar SFR surface densities than normal galaxies
- Does the gas change turbulence with older SPOGs? No, the gas stays turbulent regardless of post-burst age

What mechanism is keeping gas turbulent in SPOGs?

Page Break

How much mass is entrained in the outflows?

$$M_{\rm out} = 4.48 M_\odot \times \left(\frac{L_{\rm Ha}}{10^{-35}\,{\rm erg\,s^{-1}}}\right)\left(\frac{\langle n_e \rangle}{100\,{\rm cm^{-3}}}\right)^{-1}$$

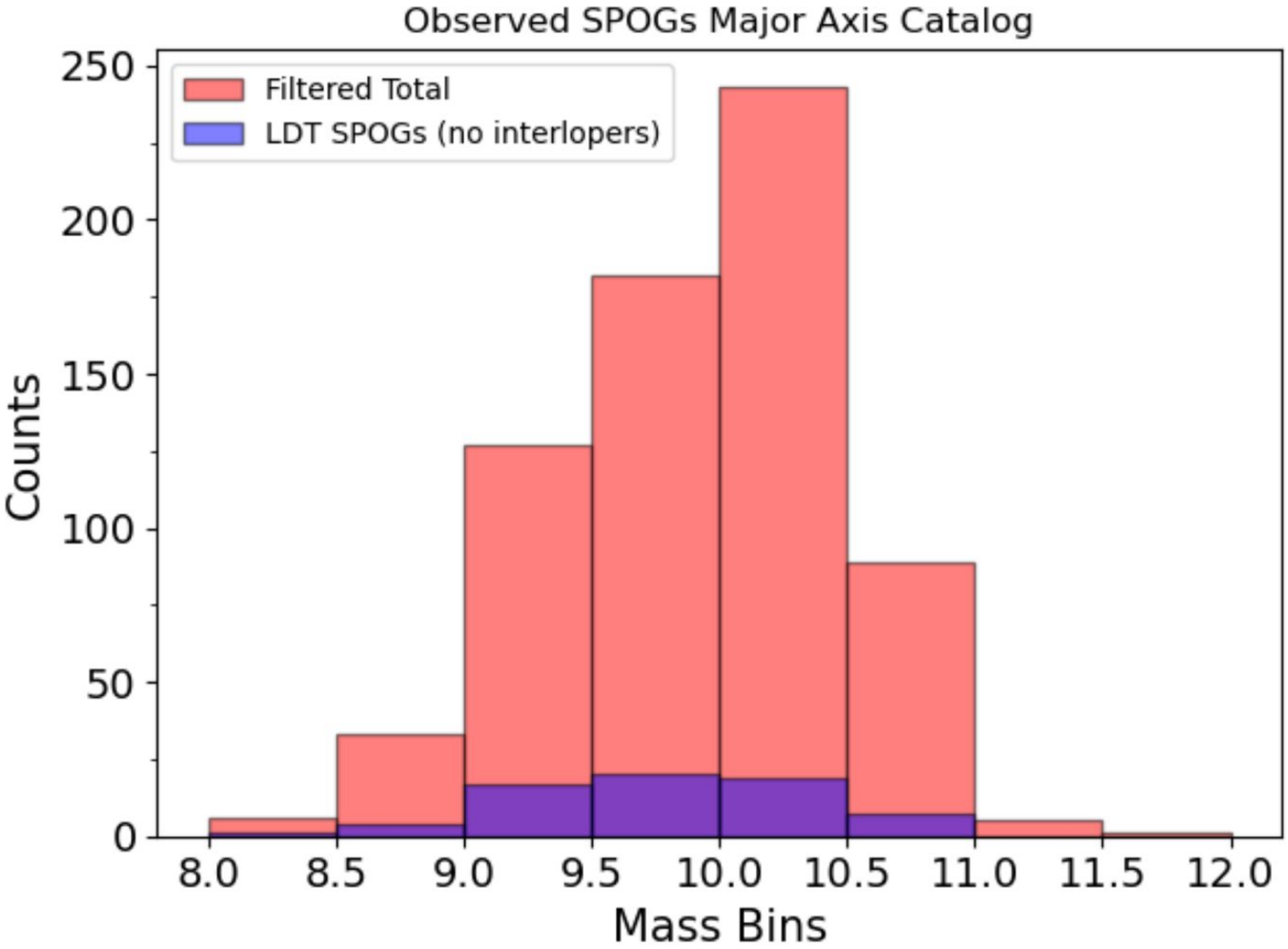
Osterbrock&Ferland+06

How efficient are the outflows?

Assuming all of the gas mass is from the recent starburst, we want to look at the outflow efficiency:

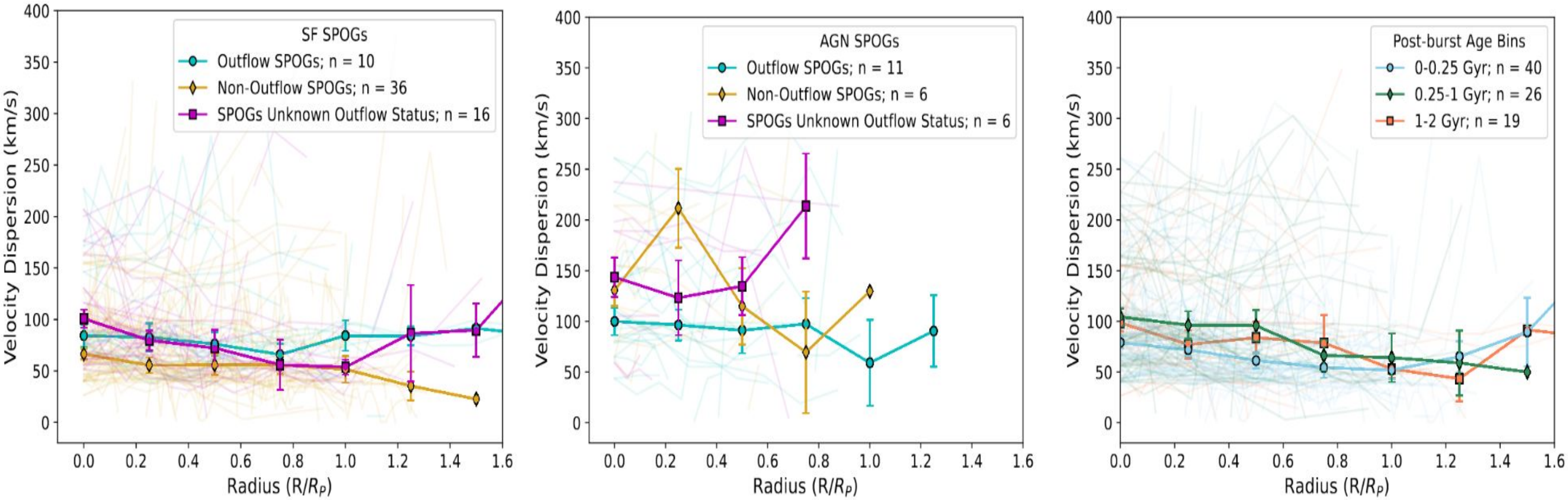
$$Mass\ loading\ factor = \frac{\dot{M}_{out}}{\dot{M}_{acc}} = \frac{\dot{M}_{out}}{\dot{M}_{SFR}}$$

Because many of our galaxies are edge-on, we need to use post-burst age for time instead



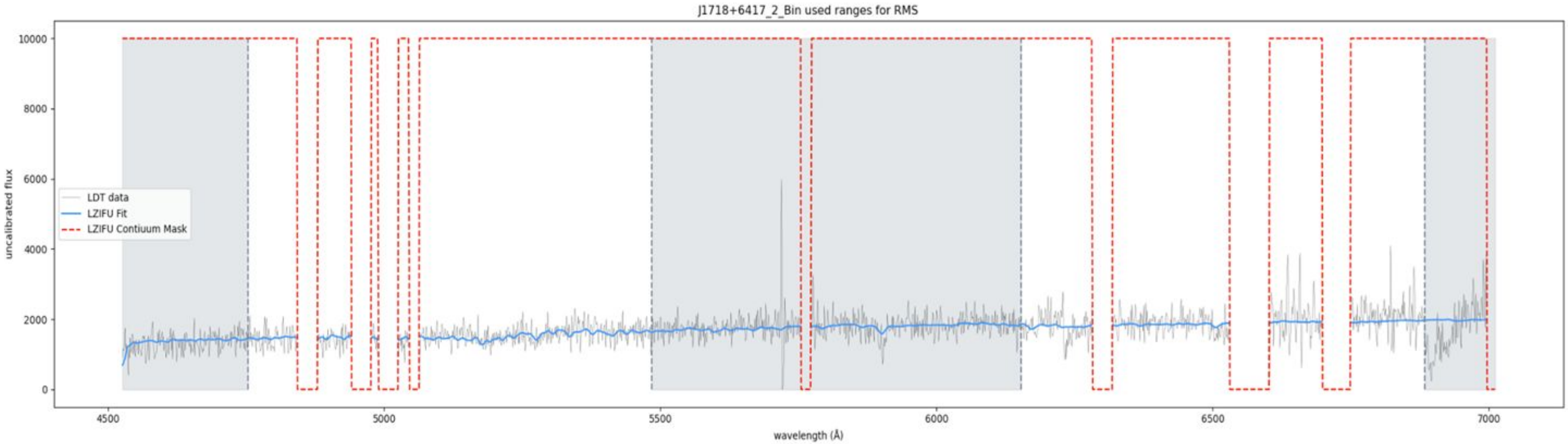
Fodor et al. in prep

We don't notice major changes in the gas



Fodor et al. in prep

Continuum color ratio as a measure of stellar population age



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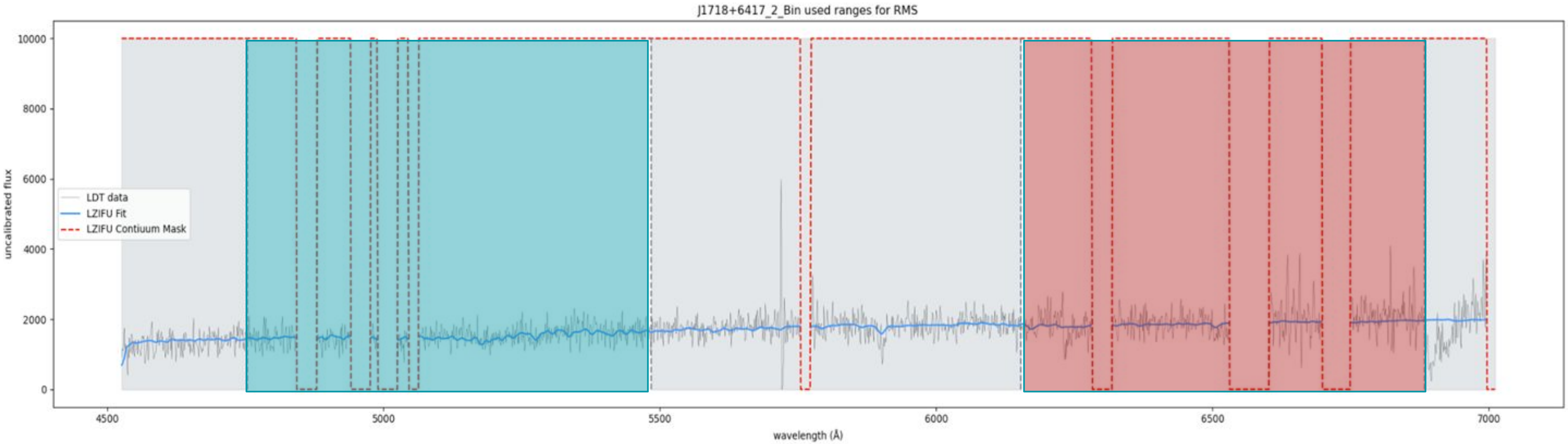

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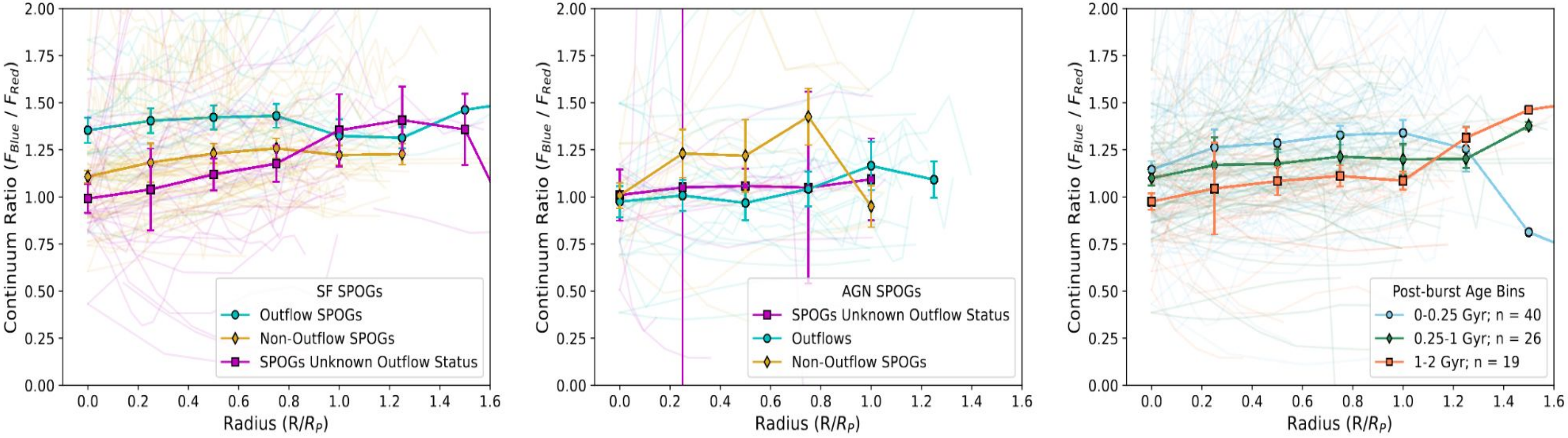

(a) (b) (c)

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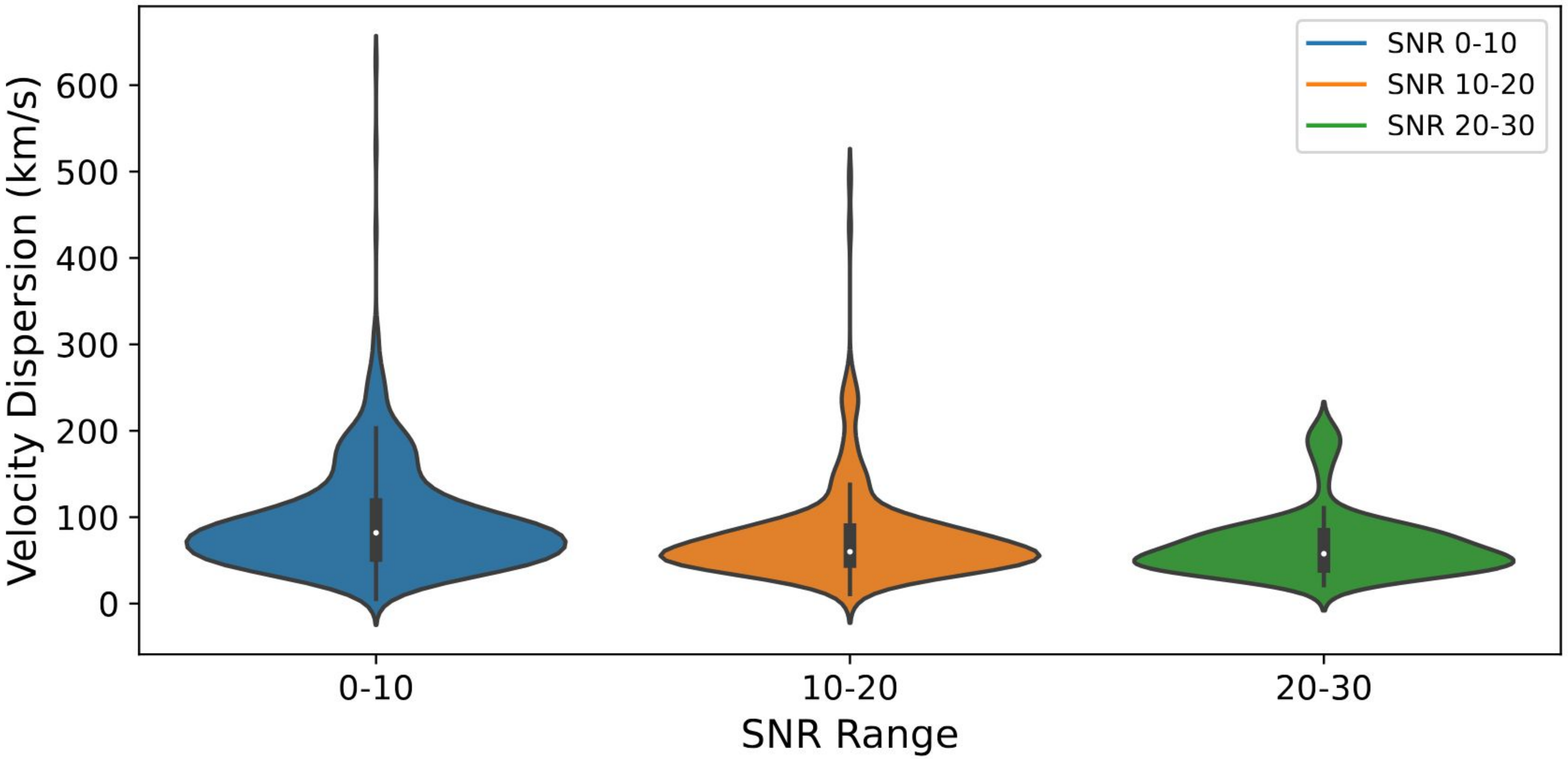
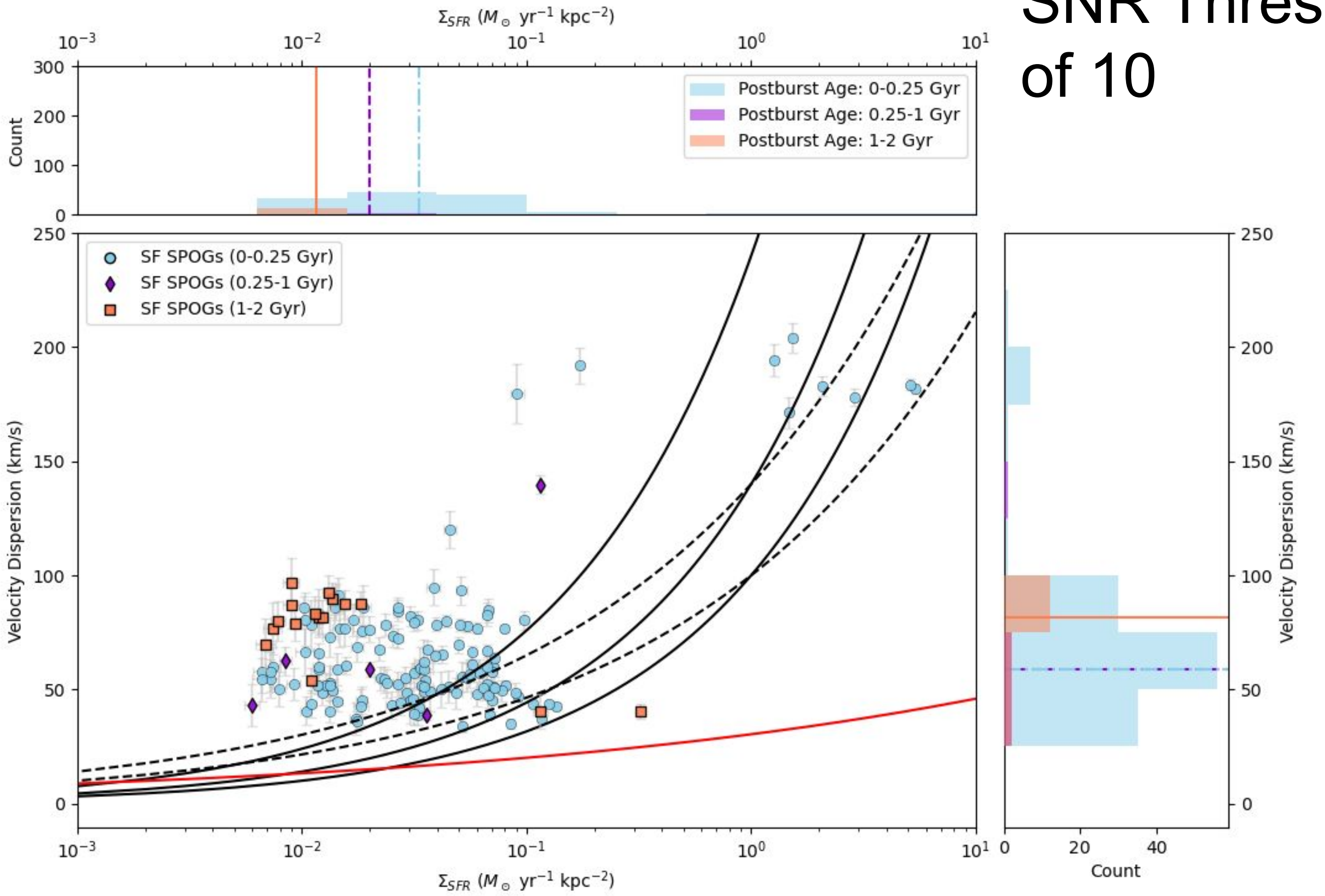
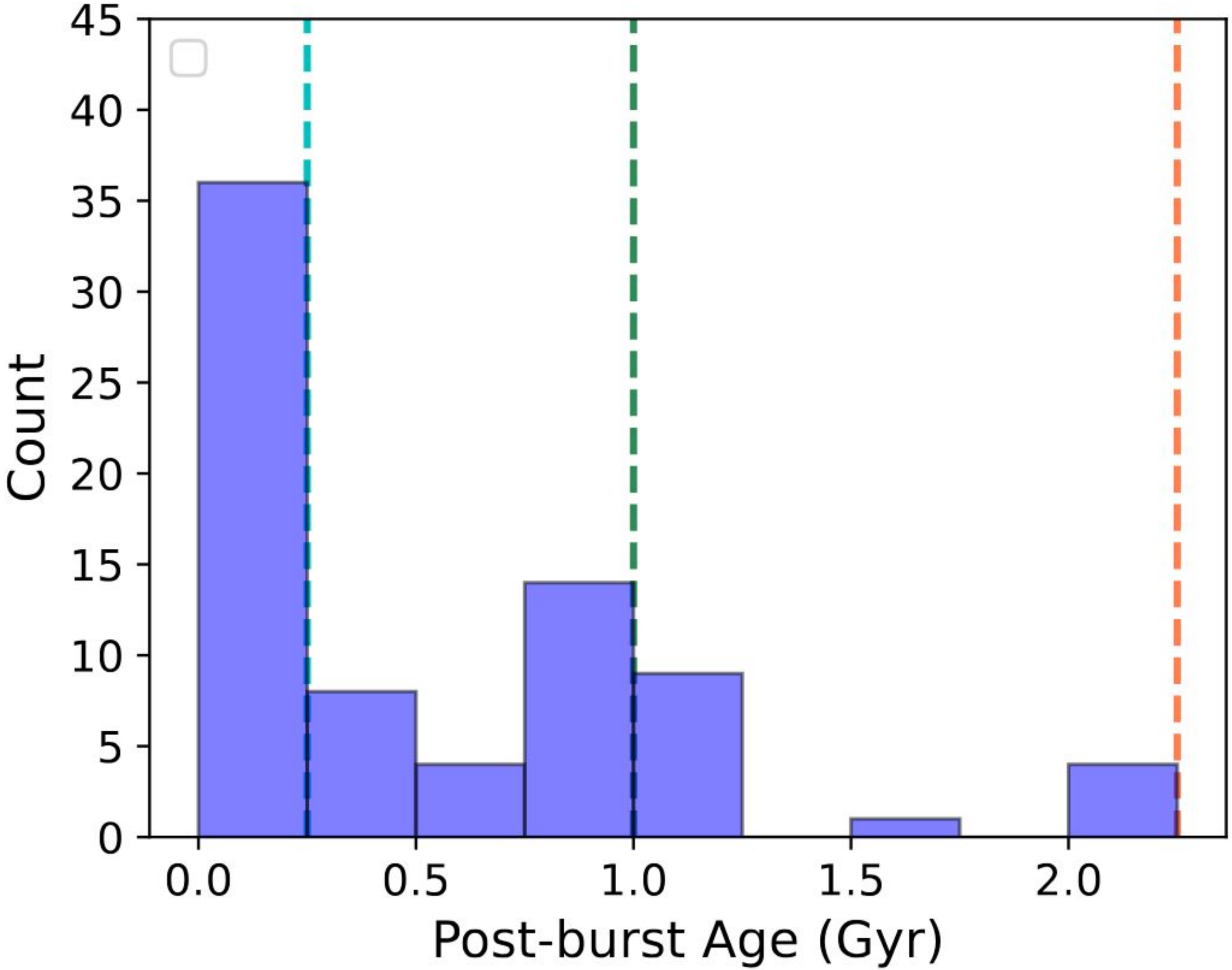
SNR of Velocity



SNR Threshold of 10





45
40
35
30
25
20
15
10
5
0
Count
0.0
0.5
1.0
1.5
2.0
Post-burst Age (Gyr)